

15/09/21

sabato 11 giugno 2022 09:17

DETERMINARE se il seguente linguaggio è CFL:

$$D = \{ a^h b^k a^l \mid k=h \text{ e } l=2h \}$$

SVLG

$$S = a^p b^p a^{2p}$$

$$1) S = \underbrace{aaaa \dots a}_p \underbrace{bbbb \dots b}_p \underbrace{aaaaaa \dots a}_{2p}$$

$vxy$

$$\left[ \begin{array}{l} v=a \\ x=a \\ y=a \end{array} \quad \begin{array}{l} v=a^+ \\ x=a^+ \\ y=a^+ \end{array} \right] \text{UP: } a^s b^p a^{2p}, \quad s \neq p, \notin L$$

stessa cosa se  $vxy$  sta nella parte  $b^p$ .

$$2) S = \underbrace{aaaa \dots a}_p \underbrace{bbbb \dots b}_p \underbrace{aaaaaa \dots a}_{2p}$$

$vxy$

$$\left[ \begin{array}{l} v=a \\ x=a \\ y=a \end{array} \quad \begin{array}{l} v=a^+ \\ x=a^+ \\ y=a^+ \end{array} \right] \text{UP: } a^p b^p a^s, \quad s \neq 2p, \notin L$$

$$3) S = \underbrace{aaaa \dots a}_p \underbrace{bbbb \dots b}_p \underbrace{aaaaaa \dots a}_{2p}$$

$$\left[ \begin{array}{l} v=a \\ x=ab \\ y=b \end{array} \quad \begin{array}{l} v=a^+ \\ x=a^+b^+ \\ y=b^+ \end{array} \right] \text{UP: } a^s a^l b^l b^s a^{2p} \rightarrow a^9 b^9 a^{2p},$$

$2p$  non è il doppio di 9.  $\notin L$

$$4) S = \underbrace{aaaa \dots a}_p \underbrace{bbbb \dots b}_p \underbrace{aaaaaa \dots a}_{2p}$$

$vxy$

$$\left[ \begin{array}{l} r=b \\ x=ba \\ y=a \end{array} \right] \circ \left[ \begin{array}{l} v=b^+ \\ x=b^+a^+ \\ y=a^+ \end{array} \right] \text{UP: } a^p b^l b^s a^s a^q, \quad a^p b^q c^m, \quad m > q$$

$$m=2q$$

$$p \times q \notin L$$